

Objectives:

Proceed with the data analysis and documentation of the experimental/simulation work.

Formulate the theoretical framework for the Power, Performance, and Area (PPA) architectural comparison.

Choose the appropriate technique to document and present the verification results with proper engineering justification.

Activities & Summary**Report Writing & Verification Documentation**

With the technical hardware validation successfully completed in Week 4, the primary focus for Week 5 transitioned to formal documentation. I began drafting the "Verification Results" chapter of the final thesis. This involved compiling the golden waveform screenshots captured via Verdi, specifically annotating the timing diagrams to visually explain the autonomous Quad Page Program (0x34) generation, the Execute-In-Place Quad Read (0xEC) fetching, and the precise nanosecond timing of the qspi_data_oen bidirectional bus turnaround.

PPA Theoretical Framework Formulation

Following the discussion with my supervisor, I began structuring the methodology for the PPA comparative analysis. I established the baseline metrics required to fairly compare three distinct memory architectures:

1. Legacy SPI with SRAM shadowing (High area, high latency).
2. A commercial-style Dual-Mode QSPI Controller (Moderate area, high software flexibility).
3. My proposed custom "Pure XIP" Hardware Accelerator (Minimal area, lowest latency). I outlined the evaluation parameters, focusing heavily on clock cycle latency per memory fetch (Performance) and the reduction of state-machine registers/multiplexers (Area).

Synthesis & Timing Analysis Preparation

To move beyond purely theoretical PPA claims, I prepared the custom Verilog RTL for logic synthesis. I organized the constraints file to extract realistic gate-count approximations and critical path timing metrics. This empirical data will be used to mathematically prove that stripping away the software-accessible MMIO registers directly reduces the overall silicon area footprint of the Pure XIP design.

Student's Name/Signature: Date:	Supervisor's Name/Signature: Date:
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Challenges & Solutions

Challenge: Establishing a fair and mathematically sound PPA comparison between my Pure XIP architecture and a commercial-style Dual-Mode controller, without expending weeks of project time to design, build, and verify a complete Dual-Mode controller from scratch.

Solution: Rather than designing a secondary controller, I chose to utilize theoretical modeling alongside empirical synthesis data. I synthesized my existing Pure XIP design to establish the absolute baseline for Area (gate count) and Performance (clock cycles per fetch). I then theoretically modeled the expected PPA overhead of a standard Dual-Mode controller by calculating the requisite architectural additions (e.g., an AHB address multiplexer, software-accessible CTRL and TX/RX_FIFO flip-flops, and the additional FSM states required for command polling). This allowed me to rigorously justify the PPA advantages of my proposed architecture without derailing the project timeline.

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WEEK 1 - WEEK 5

Summary of weekly activities

Student's Name/Signature:

Date:

Supervisor's Name/Signature:

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